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Explainable machine learning for student dropout prediction and tailored interventions in online personalized education

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Abstract
The prediction of student dropout and development of specific intervention strategies continue to be major obstacles for contemporary educational systems operating through online personalized learning platforms. This study develops explainable AI (XAI) in education framework which predicts student dropout and generates specific interventions for online personalized educational settings. This research investigates three vital elements which include (i) studying how predictive learning analytics support AI model stability and generalization performance across various educational fields and (ii) adding student learning preferences as additional contextual data and (iii) building personalized explanations for students to help teachers develop intervention strategies. We trained and tested two ensemble algorithms (Random Forest and XGBoost) with a publicly available dataset which included demographic information and learning patterns and individual preference data. To improve model transparency, we applied SHAP for student retention analysis to provide both global and instance-level interpretations of model decisions, enabling educators to understand the drivers of predicted dropout risk. The models demonstrated strong performance across courses, with the highest accuracy observed in Data Science scenarios and lower performance in Web Development. The model applied learning-style characteristics to make decisions, yet these characteristics did not produce any significant variations in student dropout rates. The research findings show that learning-style characteristics failed to produce any significant predictive results, which confirms earlier studies that questioned their usefulness in academic environments. Notably, the predictive models identify relationships between variables, but they do not show how these variables affect each other, so users need to understand these results as machine-learning-based decision support rather than following causal instructions. The dataset comes from one public database, which restricts the ability to generalize the results and needs verification through multiple institutional datasets. Future systems require specific monitoring systems for fairness protection during the deployment of demographic variables. The research data show that explainable AI systems help predict student dropouts and create data-based intervention plans, which help build equitable online learning spaces.

Keywords Explainable AI (XAI) in Education, Predictive learning analytics, SHAP for student retention, Personalized intervention frameworks, AI model stability and generalizability, Human-in-the-loop machine learning, Course-level model stability in online education

1 Introduction

Every nation requires education as its fundamental economic driver, so they allocate billions of dollars to improve educational quality [3, 36]. The United Nations [39] Sustainable Development Goal #4 drives institutions to create self-paced online learning platforms which enable students to learn independently. The combination of economic, demographic and psychological elements prevents students from finishing their courses which results in student abandonment [10, 16, 29]. The current situation represents an urgent matter which affects universities at their highest level. The advancement of computing through Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning (DL) enables researchers to detect students who face risk [18, 22, 38]. Educational stakeholders can now direct their resources toward implementing support systems because of this development [3]. Research conducted by Albahli [1] and Almalawi et al. [3] and Saqr & López-Pernas [33] demonstrates that AI and ML systems function as black boxes which create difficulties for non-technical users to operate and comprehend them.

One commonly used ML technology for solving interpretability problems is Explainable artificial intelligence (XAI) because it provides transparent decision-making insights which enable educators and administrators who lack AI expertise to use data for making informed decisions (E. Ben [8, 12, 21, 38]). Research indicates that ensemble algorithms and metaheuristic optimization methods deliver superior results for educational data analysis and dropout prediction [13, 15, 27, 40]. The research by [15] and [40] implemented Shapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) for model explainability but their research did not focus on online learning environments. The Massive Open Online Courses (MOOCs) present an optimal environment for studying student dropout patterns as they attract numerous students who frequently leave their courses [2]. Research shows that hybrid ensemble predictive models which unite traditional ML algorithms with metaheuristics that use PSO and XGBoost and AdaBoost achieve the best possible prediction accuracy results [5, 15, 37, 40]. Thus, current research investigations have focused mainly on improving predictive model accuracy instead of understanding how individual predictions work.

Tiukhova et al. [37] show that model stability and explainability assessment between different student groups and academic years is still not well-studied in academic literature. The research defines long-term stability through its ability to preserve predictive model accuracy and interpretability when using data from various academic years and different teaching methods and student demographics. Research about learning style effects on student academic achievement has produced conflicting results [34].. For example, the research by Pashler et al. [28] and Rogowsky et al. [31] revealed no significant correlation between learning styles and academic accomplishment. Likewise, Cuevas [7] demonstrated that there is no experimental proof to support teaching methods which stem from these preferred learning styles. Research studies show a small difference between their findings about style-performance relationships because El-Sabagh [9] found a weak link between learning approaches and academic results, but other studies showed students use their preferred learning methods to interact with material even when these methods do not lead to better academic results. Little has been done using machine learning to investigate and understand how student learning preferences impact model predictions during student dropout rate prediction tasks. Research in educational psychology demonstrates students learn course content through separate learning methods [7, 11, 32]. Research studies in recent years have questioned the success of learning-style based educational material design approaches. Research on algorithm transparency through global explanations exists but little has been done on creating individualized explanations for students [33]. In this paper, we investigate how XAI output enables educational institutions to inform resource allocation and policy development, moving beyond the standard accuracy metrics that typically receive most attention. Specifically, we:

- Examine the long-term stability of XAI predictive models across multiple courses.
- Analyzing the influence of learning style attributes on model decisions and feature importance rankings within the context of student dropout prediction.
- Deliver instance-specific, explicable predictions, facilitating the design of individualized interventions and student feedback.
- Develop pragmatic frameworks for translating predictive insights into workable interventions and policy guidance.

This study develops a predictive system which proves valid for student dropout prediction in online individualized education to fill existing knowledge gaps in AI applications for student dropout prediction. We establish three essential advancements for learning analytics, machine learning in education and educational technology domains:

- We investigate how predictive accuracy and machine learning model interpretability maintain their stability through time in different course settings which represent different student groups and academic periods. The study of model performance and feature relevance through Shapley Additive exPlanations (SHAP) across multiple learning environments demonstrates that AI-based dropout prediction and explanation methods remain reliable throughout different time periods and educational settings, an area previously underexplored in the literature [37].
- We also look at how student learning style characteristics fit into a dropout prediction model as contextual elements and how they are analyzed. The researchers investigate how learning styles affect predictive model decision-making through XAI methods while understanding their role as indicators of general learner characteristics that influence risk evaluation. Thus, acknowledging the continuing controversy over the direct instructional validity of learning styles [28, 31].
- Third, we underline SHAP's use and creation of instance-level explanations. Beyond aggregated model transparency, we show how personal feature attributions can provide a detailed analysis of the specific elements influencing each student's projected dropout risk, directly empowering teachers to create and customize focused interventions and support plans for individual students.
- Fourth, we suggest a pragmatic intervention framework that closes the gap between actionable educational practice and predictive model output. This framework provides a clear way for institutions to operationalize predictive analytics to improve student retention by converting model-generated dropout probabilities and instance-

level explanations into a systematic process for identifying high-risk students and guiding flexible support delivery.
- Finally, our study utilized ensemble machine learning models (Random Forest and XGBoost) on a publicly available dataset. This provided empirical evidence supporting the feasibility, performance, and interpretability of the comprehensive framework for student dropout prediction in online personalized education.

The combined research activities enhance both theoretical understanding and practical implementation of explainable artificial intelligence systems in educational environments. The development of dependable systems with interpretable features for student achievement support and reduced student dropout rates remains a priority. The remaining sections follow this structure: Sect. 2 reviews existing research before presenting the study methodology in Sect. 3. The paper presents its results and discussions in Sect. 4 followed by conclusions and future work in Sect. 5

2 Related works

Factors that influence students' academic performance are multidimensional in nature [21, 27], from demographic and socioeconomic to psychological tendencies, behavioral involvement, learning styles, motivation, disposition, learning challenges, or even learning preferences and institutional influences. A wide range of factors shape academic achievement, so that is what defines it. Including these many elements into prediction models and instructional programs will help provide students who are at risk of poor performance or dropout with thorough knowledge and more focused support. Figure 1 presents a list of elements routinely underlined in literature [1–3, 5, 23–26, 33, 36] as affecting students' academic success.

2.1 Classical machine learning application in academic performance prediction

Nti et al. [24] developed a predictive system which employed two traditional machine learning approaches to forecast student academic results through their social media activities. The research findings demonstrated that students who use social media during class hours will achieve lower academic results. The research findings remain restricted because the study only examined one university with a small participant group. Similarly, Nti & Quarcoo [25] demonstrated that students who demonstrate strong self-confidence and positive thinking and self-efficacy achieve better results than students with lower levels of these traits in programming education. Likewise, the research by Owusu-Boadu et al. [26] applied multiple classical machine learning classifiers to study which academic and non-academic elements influence student achievement levels. The study results from a particular location could be enhanced through the addition of more diverse datasets.

2.2 Explainable artificial intelligence application in academic performance prediction

Tiukhova et al. [37] studied XAI potential to boost student achievement prediction through model stability assessment across different academic settings. The research applied self-regulated learning theory together with existing predictive analytics findings to achieve its goals. The researchers established course-level attributes through concept drift detection methods to study model performance variations between different student groups. The researchers employed SHAP to determine which learning activities

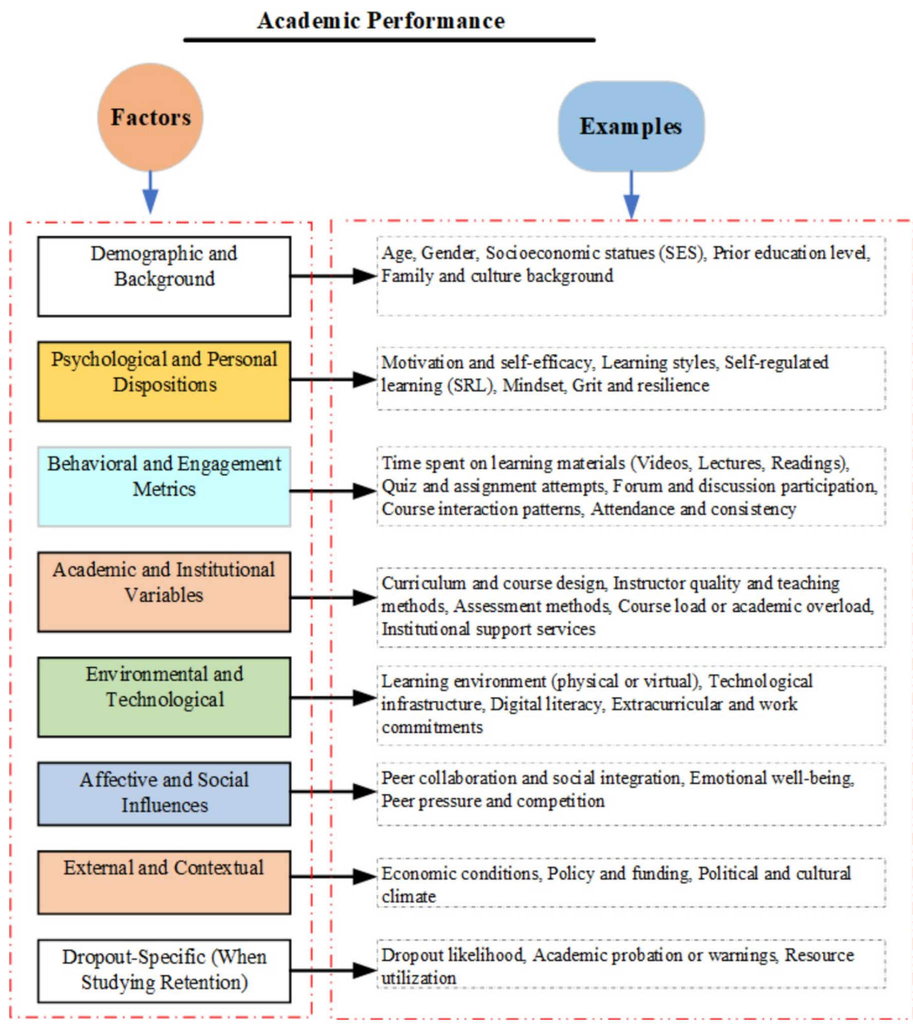

Fig. 1 Conceptual categorization of factors influencing students' academic performance

produced substantial effects in their analysis. The research Tiukhova et al. [37] shows that the value of some features changes from subject to subject, but it also shows that there are stable predictive patterns that can be used to create personalized therapies. The research enhances comprehension and the practical significance of learning analytics studies. Nagy and Molontay [21] examined XAI-based interpretability for predicting student dropout rates in higher education institutions. The authors utilized CatBoost and several machine learning models to swiftly identify pupils with heightened failure risk. The authors utilized three distinct methodologies, including permutation importance, partial dependence plots, and SHAP values, to elucidate model interpretations on a bigger scale. The study was limited to a single technical university, necessitating the application of similar methodologies in various higher education institutions in future research. To fully explain student success, researchers need to look at both pre-enrollment achievement data and socioeconomic and psychological factors. Goran et al. [13] conducted study on the use of AI systems in addressing the significant issue of student attrition from higher education institutions. The study integrated Adaptive Boosting (AdaBoost) and eXtreme Gradient Boosting (XGBoost) machine learning models with metaheuristic optimization strategies to enhance the accuracy and reliability of dropout prediction systems. The researchers developed SCHO as an innovative optimization technique to address current metaheuristic challenges. The new version is better at finding pupils who are at danger of failing in school. The research employs explainable AI techniques, such as SHAP and SAGE, to ascertain the reasons for student attrition. The analysis enables educational staff to direct their support toward maintaining student enrollment. The proposed solution faces performance limitations because it depends on limited computational resources which hinder optimization operations. Likewise, Parkavi et al. [27] developed a decision-support system using swarm intelligence and explainable artificial intelligence methods to evaluate student academic achievement systematically. Using Henon execution and chaotic particle swarm optimization, a metaheuristic approach superior to genetic algorithms and ant colony optimization is presented [27]. The study confirms the significant influence of ICT on student self-progress using Spearman's rank correlation. In conclusion, the study recommended further research in practical learning environments and comparative studies using different optimization strategies. Bulut et al. [5] aimed to demonstrate how human–machine collaboration can improve the accuracy and speed of high school educational decision-making by examining dropout prediction as one example. Findings demonstrated that random forest outperformed deep learning. It also showed how important actionable variables are, such as students' GPA, sense of belonging at school, and belief in their own abilities in math and science, which can influence students' performance. Also, the random forest and deep learning models recorded a low recall of 0.17 and 0.21, respectively. This means that their ability to find failure cases was lower. We believe this may be due to a significant difference in the goal variable across classes in the publicly available education data source used for ML analysis. Again, the differences between LIME, breakdown, and SHAP analysis in this study show how important it is to examine data from different XAI techniques, as they may reveal subtle and useful information.

Kala et al. [15] developed a hybrid deep learning system to predict student performance in traditional classroom settings through the combination of particle swarm optimization (PSO) and deep neural networks (DNN). The authors used a detailed dataset that combined student demographic data and academic performance and high school records and higher education admission test results to predict student success in programming courses. The research demonstrated that the PSO-DNN model achieved superior results than all other deep learning and machine learning approaches. The research faces its main restriction because it studies a particular educational institution and academic program which reduces the ability to apply its results to other settings. Shoaib et al. [35] developed an AI student success prediction system which transforms conventional campus management systems (CMSs) through its innovative approach. The authors spent extensive time building a unified database by uniting student data from multiple relational databases. The researchers developed a convolutional neural network (CNN) feature-learning block to detect hidden data patterns and an ensemble classification model which combines SVM with random forests and KNN classifiers and Bayesian averaging for improvement. The system achieves high precision in forecasting student academic performance and risk assessment and student retention and dropout behavior. The research study faces two main restrictions because it operates within a single university setting and requires additional testing across different educational environments. Saqr and López-Pernas [33] used XAI techniques to study how personalized instance-level forecasting works. The research investigated how incorrect predictions affect feedback systems and classroom decision-making processes. The researchers applied five machine learning algorithms to find random forests produced the best results and used SHAP to explain individual prediction outcomes. The research demonstrates that explainable artificial intelligence provides understanding of essential factors yet maintains separate domains for algorithmic predictions and educational concepts. The research demonstrates that human-AI collaboration stands as an essential requirement for success. The authors understand that online data has its restrictions and human learning processes contain multiple complex elements. The authors suggest that additional detailed findings could emerge from extensive research that includes personal student data.

Albahli [1] conducted an extensive evaluation of machine learning approaches which predict academic results at Saudi universities. The author performed data preprocessing followed by feature engineering before evaluating RF and K-Nearest Neighbors and CNNs as prediction models. The research findings show that CNNs achieve superior performance compared to other machine learning approaches. The research established which social and demographic factors enhance prediction accuracy. The research faces two main restrictions because it studies only Saudi institutions and lacks historical data which reduces the study's applicability to other contexts. The evaluation of student performance requires additional variables including socioeconomic status and psychological factors to achieve complete understanding. E. Ben George [8] developed an XAI system which used random forest classification to predict student academic results. The research achieved better results but suggested future work should focus on adding student behavioral characteristics to model training. The research did not evaluate how the model performs when analyzing different academic subjects. Geethanjali and Umashankar [12] investigated how XAI systems help predict student academic results. The research established that educational AI systems need complete acceptance from all educational stakeholders including students and teachers and parents and government officials [12]. Ujkani et al. [38] used multiple machine learning and deep learning approaches to develop student academic performance prediction systems. The researchers used SHAP as their XAI method to reveal which factors determine student success or failure. The research findings show high accuracy, but the authors suggest that student behavioral characteristics impact the success of AI-based educational interventions.

The research findings from these studies demonstrate that explainable AI (XAI) plays a vital part in enhancing student achievement forecasting systems and dropout prevention methods which operate in different educational settings. The research by Tiukhova et al. [37] shows that learning attributes which remain stable throughout time continue to matter even when educational environments shift because of COVID-19. The research by Nagy and Molontay [21] shows that explainable instance-level feedback holds promise for better student results, but they support using multiple data sources that extend past academic performance metrics. The research by Goran et al. [13] and Parkavi et al. [27] shows that ensemble methods with swarm intelligence optimization produce better predictions but researchers need to test these methods in actual educational settings for practical implementation. The research by Shoaib et al. [35] and Kala et al. [15] uses deep learning with XAI to achieve better model interpretability and performance, but they identify constraints from institutional settings and specific courses. The research by Bulut et al. [5] demonstrates that human-AI collaboration through hybrid systems solves data imbalances to enhance prediction accuracy. The research findings support this study through their emphasis on model transparency and longitudinal data analysis and personalized interventions and their ability to perform well in multiple educational settings.

3 Research approach

The research design used a systematic method which consisted of multiple stages. The research method combines exploratory data analysis (EDA) with predictive modeling and longitudinal stability analysis and explainable AI (XAI) techniques. The paper explains each research phase in detail throughout its sections.

3.1 Data collection and preprocessing

The study data originated from Kaggle which serves as a leading platform for researchers to access free public datasets. The dataset consisted of 10,000 entries which included 15 different features. The dataset contains student_ID as well as age and gender and education_level and course_name and time_spent_on_videos and quiz_attempts and quiz_scores and forum_participation and assignment_completion_rate and engagement_level and final_exam_score and learning_style and feedback_score and dropout_likelihood. We checked the dataset for data inconsistencies which included checking for missing values and outliers and inconsistent numerical value ranges. The data set contained no missing entries. The one-hot encoding technique transformed categorical data points (education level, gender, learning style, course name and engagement level) into numerical values for machine learning model processing efficiency. Most machine learning algorithms function optimally when dealing with numerical data instead of categorical information. The min–max normalization technique (Eq. 1) transformed numerical study data features (time spent on videos and quiz scores) into comparable values. The clean dataset $DS(X, y)$ was partitioned into 80% for training and 20% for testing.

$$x_{norm} = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

The study data was highly imbalanced. The "Dropout_Likelihood" variable (encoded as "Yes" = 1, "No" = 0) had fewer dropouts (1) relative to non-dropouts (0). We applied the Synthetic Minority Over-sampling Technique (SMOTE) for Numeric and Categorical data (SMOTEC) as part of our data processing pipeline to address this class imbalance. We used SMOTEC to make synthetic examples of the minority class after cleaning and merging all the important features, like how engaged students were (time spent on videos and quizzes), their demographics (age, gender, and level of education), and their learning style, into matrices for the independent variables (X) and the target. The SMOTEC was used to balance the dropout (1) and non-dropout (0) classes in the study dataset. This method effectively mitigated the skew in our target variable. This ensured that our adopted machine learning models had sufficient positive-class samples to learn from. This ultimately led to an enhancement in both predictive performance and model reliability. We applied recursive feature elimination [14] to the study training dataset before actual model training to evaluate the influence of our input features. We observed that features like "education_level" and "engagement_level" were consistently among the top predictors. This initial filtering process aimed to ensure that the most important features were the focus of the predictive models.

3.2 Predictive modeling

The predictive modeling task was formulated in this paper as a binary classification problem, as defined in Eq. 2, where Y indicates the dropout likelihood (0 = No dropout, 1 = Dropout). Thus, given input feature vector $x \in R^d$, the goal is to learn a function f: $R^d \rightarrow \{0,1\}$ that predicts the dropout label $y \in \{0,1\}$. To accomplish this, two widely used [1–3, 5, 23–26, 33, 36] machine learning algorithms were selected, namely Random Forest and Extreme Gradient Boosting (XGBoost). The grid search algorithm was used to tune all model hyperparameters in this study.

$$Y \in \{0,1\} \tag{2}$$

Random Forest [4] is an ensemble machine learning technique. It constructs multiple decision trees on random subsets of the data and features. Each tree independently outputs a class prediction, and the ensemble's final decision was determined by majority voting in this study. Let DT be the training dataset of size n. For each tree T_k, we sample DT_k from DT with replacement. We then learn a decision tree T_k on DT_k by recursively splitting the feature space as defined in Eq. 3. In classification, impurity is often the Gini index or entropy. The forest aggregates predictions from K trees as defined in Eq. 4. The out-of-bag sample technique was used in this study to assess RF ensemble stability. The model's predictions were subject to XAI techniques (SHAP) to examine the contribution of dataset features such as "learning_style," "time_spent_on_videos," etc.

$$Nodesplitcriterion = argmin_{j\theta}\left[Impurity\left(node\right) - \left(Impurity\left(leftsplit\right) + Impurity\left(rightsplit\right)\right)\right] \tag{3}$$

$$\hat{y}RF = MajorityVote\left(T_1\left(x\right), T_2\left(x\right), .., T_K\left(x\right)\right) \tag{4}$$

XGBoost [6] is a boosting algorithm. It iteratively refines a strong learner by adding decision trees to minimize a given loss function. Each new tree attempts to correct the errors from the preceding ensemble of trees. Let $\hat{y}^{(m-1)}\left(x\right)$ be the prediction from the $(m-1)^{th}$ iteration. The m^{th} tree f_m is fitted from the pseudo-residual derived from the objective function. For iteration m, the pseudo-residual for the sample i is defined as in Eq. 5.

$$r(m) = \frac{\partial}{\partial \hat{y}_i} L(y_i, \hat{y}_i)\,|\hat{y}_i = \hat{y}_i(m-1) \tag{5}$$

where $L\left(y_i, \hat{y}_i\right)$ is the loss function, say logistic loss for a classification task, $\hat{y}_i^{(m-1)}$ is the predicted score for the i^{th} instance at iteration $(m-1)$. We took the negative gradient of the loss with respect to the predicted score $\hat{y}_i$. This gives the direction in which the model should update its prediction to reduce the loss for the next iteration. Once the pseudo-residuals are computed. The algorithm fits a new tree f_m to the residuals. The model is then updated as defined in Eq. 6., where $\hat{y}^{(m)}\left(x\right)$ is the updated prediction for a given input x after m boosting rounds.η is the learning rate, scales the contribution from the new tree, and $f_m\left(x\right)$ is the newly trained tree.

$$\hat{y}^{(m)}(x) = \hat{y}^{(m-1)}(x) + \eta \cdot f_m(x) \tag{6}$$

Like the RF, the final predictions are validated on a holdout set. SHAP was used to reveal how individual features contribute to the predicted dropout probabilities at both a global (feature importance) and local (instance-level) scope.

3.3 Explainable AI (XAI) techniques

We implement XAI methods to ensure interpretability of our model's predictions and help propose an intervention strategy for stakeholders. The Shapley Additive exPlanation (SHAP) [17] provides consistent, additive explanations at both global and instance levels. Equation 7 defines the SHAP, where $f\left(x\right)$ is the prediction, ϕ_0 is the baseline prediction, and ϕi is the contribution of the feature i.

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i \cdot x_i \tag{7}$$

The Local Interpretable Model-agnostic Explanations (LIME) [30] as defined in Eq. 8, was used to provide individual prediction explanations through local model approximations. The model maintained its stability throughout different courses through performance drift analysis. The study used course names instead of semester or year to establish its cohort definitions. The model stability metric is defined in Eq. 9. The performance difference between two separate courses serves as the measurement for this study. The study employed Spearman's rank correlation defined in Eq. 10 was adopted to assess how well SHAP value explanations maintained their stability between different courses.

$$\text{explanations}\left(x\right) = \text{argmin}_{g \in G} L\left(f, g, \pi_x\right) + \Omega\left(g\right) \tag{8}$$

$$\text{Model Stability} = \left|\text{Performance}_{\text{course (B)}} - \text{Performance}_{\text{course(A)}}\right| \tag{9}$$

$$\rho = 1 - \frac{6\sum_{d_i^2}}{n(n^2-1)} \tag{10}$$

Where d_i^2 is the rank difference in feature importance across courses.

3.4 Model evaluation

There are several metrics for evaluating and validating a classification model. We use five metrics to evaluate the robustness of our predictive models via fivefold cross-validation. The performance metrics included: precision (Eq. 11), recall (Eq. 12), F1-score (Eq. 13), accuracy (Eq. 14), and ROC-AUC (Eq. 15).

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{12}$$

$$\text{F1} - \text{score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} \times \text{Recall}} \tag{13}$$

$$\text{Acccuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{14}$$

$$\text{AUC} = \int_0^1 \text{TRP}(\text{FPR})\text{dFPR} \tag{15}$$

where: TP = True Positives (correctly predicted positives), TN = True Negatives (correctly predicted negatives), FP False Positives (incorrectly predicted positives), FN = False Negatives (incorrectly predicted negatives), and TPR (FP R) = True Positive Rate as a function of the False Positive Rate.

We included demographic information about students including their gender and educational background to understand how different population groups learn. The model's performance across different subgroups becomes a potential fairness issue when it depends on demographic information. We performed multiple diagnostic tests to assess fairness in the model. The model was evaluated to determine whether performance levels were comparable across student groups during predictive assessment. The model produced unequal numbers of false positive and false negative results which could lead to unfair recommendations for student interventions. The SHAP analysis revealed if the model used different feature weights for different demographic groups instead of depending on student behavior. These steps helped us identify where fairness concerns may arise and highlight the importance of developing future strategies to reduce demographic bias and ensure equitable model-supported decisions for all students.

4 Results and discussions

4.1 Descriptive statistical analysis of study dataset

Table 1 shows the fundamental statistical analysis of our study data. The statistical results demonstrate that participants in the study belong to different learning categories. The study found participants aged between 15 and 49 years old with an average age of 32 years while their video watching duration spanned between 10 and 499 min. The Quiz_Attempts statistics shows students need an average of 2.5 attempts to complete their quizzes. The quiz scores between 1 and 4 show students have achieved different levels of course mastery while their quiz performance averages at 64.58%. The forum participation data shows students either do not participate at all or they actively engage with the forum content. The assignment completion rate (69.55%) and final exam score (64.70%) show a wide range of performance from 40% to almost 99%. Most students show average academic performance, but many students achieve better or worse results than the average. The feedback scores distribute across three points on a 1–5 scale which shows students have average satisfaction, but their opinions range from strong dissatisfaction to complete satisfaction. The large score and behavioral variations in Table 1 make ensemble models like Random Forest and XGBoost suitable for analysis because they excel at handling diverse data types.

Figure 2 displays study data correlation matrix plot. The correlation matrix supports the findings we previously discovered. The correlation matrix demonstrates weak to strong linear relationships between all numerical data points. Most correlation values approach zero which indicates that the variables show no direct relationship with each other. The age variable shows no statistical connection with time_spent_on_videos ($r = -0.013$) or final_exam_score ($r = -0.005$). The quiz_attempts data shows minimal

Table 1 Basic descriptive statistics of study variables. The table presents a summary of key dataset features through its display of student age and video time and quiz attempts and quiz scores and forum participation and assignment completion rate and final exam score and feedback score. The table shows the following statistics for each variable: count and mean and standard deviation (std) and minimum (min) and maximum (max) and 25th percentile and 50th percentile (median) and 75th percentile

	Age	Time_Spent_on_Videos	Quiz_Attempts	Quiz_Scores	Forum_Participation	Assignment_Completion_Rate	Final_Exam_Score	Feedback_Score
count	10,000	10,000	10,000	10,000	10,000	10,000	10,000	10,000
mean	32.14	255.38	2.50	64.58	24.62	69.55	64.70	3.01
std	10.06	141.66	1.12	20.29	14.33	17.36	20.10	1.42
min	15	10	1	30	0.00	40	30	1
25%	24	131	1	47	12	54	47	2
50%	32	257	3	65	25	69	65	3
75%	41	378	4	82	37	85	82	4
max	49	499	4	99	49	99	99	5

linear connection to both forum_participation (r = –0.014) and quiz_scores (r = –0.006). The performance indicators assignment_completion_Rate and Final_Exam_Score show only a slight correlation of 0.014. The weak connections between academic results and student actions indicate that multiple hidden elements influence student completion rates in this dataset. The near-zero correlation values indicate that multiple small or hidden variables work together to determine student completion or dropout rates.

Our results show that we need to use advanced modeling methods that can handle non-linear data for analysis. The descriptive statistics show how pupils spread out based on how they do in their engagement activities and their performance scores. The variables in this dataset show almost no association, thus you can't use just one attribute, like time_spent_on_videos or feedback_score, to guess how well a student will do or how likely they are to fail. To find out what factors have the biggest impact on student performance in diverse situations, it is important to apply advanced machine learning algorithms and explainable AI (XAI) methodologies. The weak association between variables enables individualized approaches because many minor indications give greater information about how students are doing, which makes it easier to create personalized intervention plans.

4.2 Overall model performance

The performance results for both baseline and proposed ensemble models appear in Table 2. The Random Forest and XGBoost models performed better than Decision Tree and Logistic Regression in all evaluation metrics. The Random Forest model obtained 0.788 accuracy and 0.798 precision, 0.773 recall and an F1-score of 0.785. The XGBoost model produced results of 0.80 accuracy and 0.790 precision and 0.816 recall and 0.785 F1-score. The results from Table 1 demonstrate that both models performed at a similar level. The main distinction between these models exists in their ability to handle precision-recall tradeoffs because this affects real-world educational intervention methods. The XGBoost model produced a higher recall value of 0.816 compared to the random forest model which reached 0.773. The XGBoost classifier demonstrated superior performance in detecting students who would drop out because it identified more actual dropouts. The model allows teachers to detect additional students who might need help.

The XGBoost model achieved a precision rate of 0.790 which was lower than Random Forest at 0.798. The XGBoost classifier correctly identifies most at-risk students but it incorrectly identifies 21% of students who do not need intervention. The false positive results from this model will direct educational resources to wrong students which results in inefficient resource distribution. The random forest classifier achieves better precision at 0.798 which results in fewer incorrect positive flags that amount to 20.2% of its total positive predictions. The random forest classifier achieves better precision but its lower recall rate of 0.773 indicates it might fail to detect more students who could leave school. The model could fail to detect students who need help at the right time because of this approach. The institutional decision-making process requires selecting between two options which involve either identifying more at-risk students through XGBoost or reducing false positive alerts through random forest. The evaluation of support resource costs for false positive students' needs to be compared to the advantages of detecting actual dropouts and providing early assistance. The evaluation of model performance through confusion matrices (Fig. 3) supports the observed results. The Random Forest

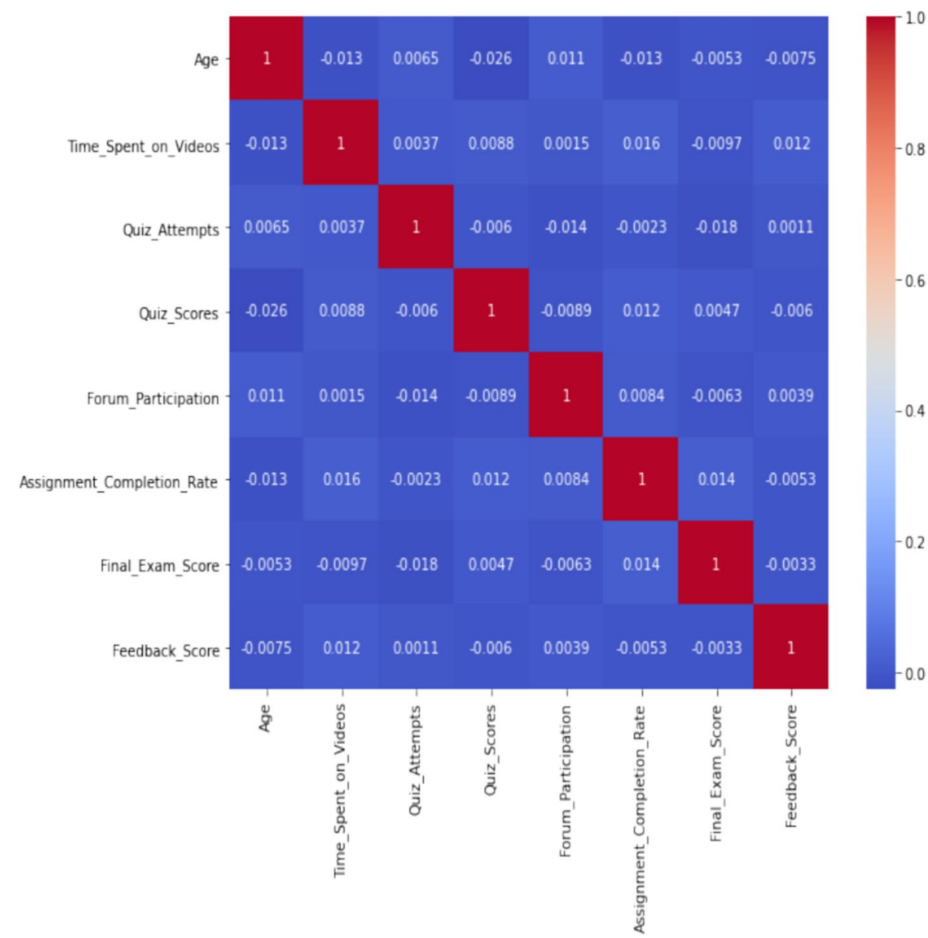

Fig. 2 Correlation matrix of key student features

Table 2 Overall model performance metrics. The table evaluates the test dataset performance of Random Forest, XGBoost, Decision Tree and Logistic Regression models through five evaluation metrics which include accuracy and precision and recall and F1-score and area under the ROC curve (AUC)

Model	Accuracy	Precision	Recall	F1-Score	AUC
Decision Tree	0.678	0.665	0.717	0.785	0.678
Logistic Regression	0.654	0.647	0.675	0.785	0.654
Random Forest	0.788	0.798	0.773	0.785	0.788
XGBoost	0.800	0.790	0.816	0.785	0.800

model correctly identified 1,940 students who would not drop out but it incorrectly predicted 473 students would leave school. The XGBoost model achieved better results by identifying 1,968 students who would drop out and missing only 445 students who left school. The models demonstrate strong performance according to their evaluation results which supports previous research about ensemble-based methods in different learning settings and data types.

The performance of our models was compared with two other baseline models. Namely, decision tree and logistic regression. The decision tree model achieved 67.8% accuracy and an AUC of 0.678. We observed an accuracy of 65.4% and an AUC of 0.654 by the logistic regression model. In contrast, our ensemble models (Random Forest and

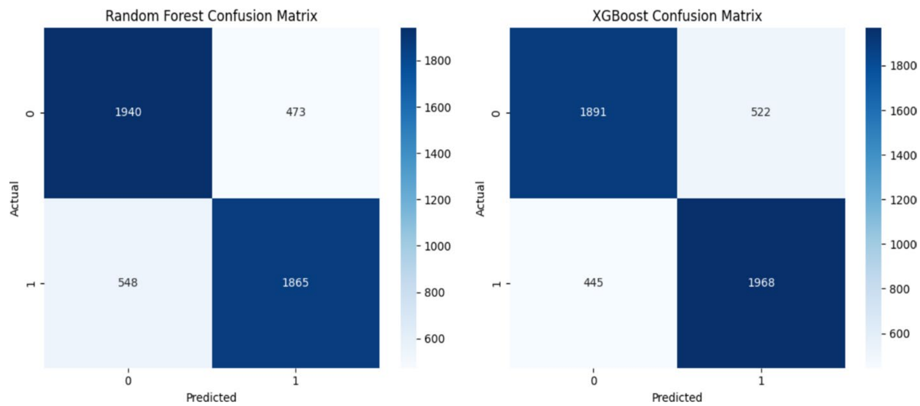

Fig. 3 Test dataset confusion matrices for Random Forest (left) and XGBoost (right) models. The matrices show the number of correct and incorrect predictions for each model through their true positive and true negative and false positive and false negative counts. The XGBoost model shows better performance than Random Forest because it achieves higher true-positive and true-negative rates which results in better recall and predictive accuracy

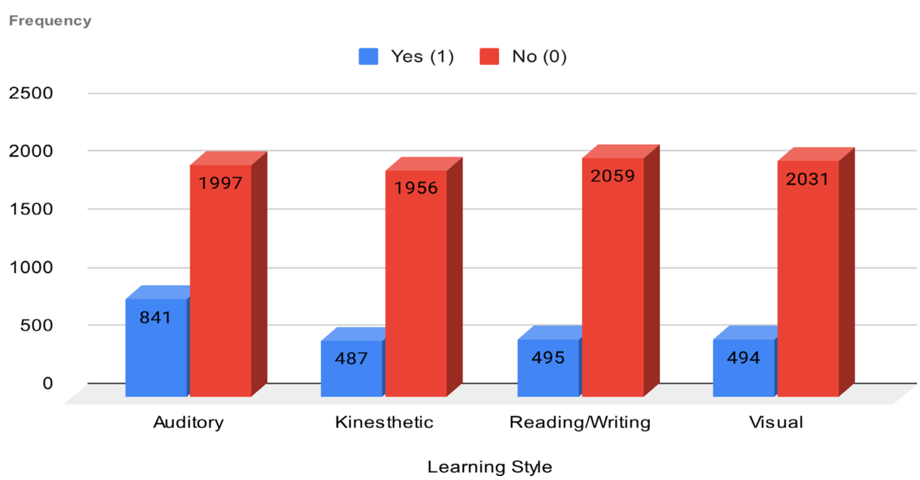

Fig. 4 Distribution of dropout likelihood by learning style. The graph shows how students who dropped out (Yes = 1) compared to those who stayed (No = 0) within four learning style groups: Auditory, Kinesthetic, Reading/Writing and Visual. The dropout rates between learning styles show minimal differences which indicates that learning style does not effectively predict student dropout behavior

XGBoost) achieved significantly higher accuracies (78.8% and 80.0%, respectively) and AUC scores (~ 0.80). This result demonstrates the advantage of combining advanced ensemble methods with explainable AI for dropout prediction. It should be noted that the outcomes reported in this study reflect predictive associations rather than causal relationships.

The crosstab plot in Fig. 4 demonstrates how student learning styles relate to their dropout status. The crosstab analysis shows that dropout rates (1) and non-dropout rates (0) distribute equally across all four learning styles (auditory, kinesthetic, reading/writing and visual). The Reading/Writing style contained more students in total but the distribution of students between non-dropouts and dropouts remained similar across all four styles. The dropout rate for auditory learners reached 19% because 1,997 students did not drop out while 481 students did. The dropout rate for reading/writing learners reached 19.4% at the same level as other groups. The visual and kinesthetic groups demonstrate identical dropout patterns which show no significant variation in their dropout rates.

The similar dropout rates between different learning style groups indicate that students' learning preferences do not effectively predict their likelihood of dropping out. The implementation of style-specific learning promotion programs will unlikely lead to substantial decreases in student withdrawal rates. The recognition of student learning diversity according to Fleming and Mills [11] remains essential even though their preferences do not directly link to dropout rates. The diverse learning preferences students exhibit according to Fleming and Mills [11] and Zain [20] require educators to create multiple learning approaches and adaptable educational routes. Educational content that enables students to select their preferred learning approach through interactive labs and audio lectures and visual aids and text summaries effectively supports different learning preferences. The strategic implementation of learning materials that match individual student preferences helps teachers create better engagement with their students.

4.3 Course-by-course model performance and long-term stability analysis

Before assessing how well our suggested framework works in diverse educational settings, the study team needs to make sure that it can accurately predict outcomes. We did a performance review for each course. Figures 5 and 6 show that the random forest model and the XGBoost model are both stable over time when used with different courses. The research findings demonstrate that the framework operates consistently across several courses, hence validating its compatibility with diverse online learning systems. The assessment of five courses indicated that random forests and XGBoosts yielded comparable outcomes in terms of accuracy, F1 scores, and AUC values. The findings indicate that dropout prediction models yield consistent outcomes across various educational contexts. The model's performance varies across courses, indicating that elements from each course influence the model's predicted accuracy. The investigation shows that algorithms work differently depending on three important things: the

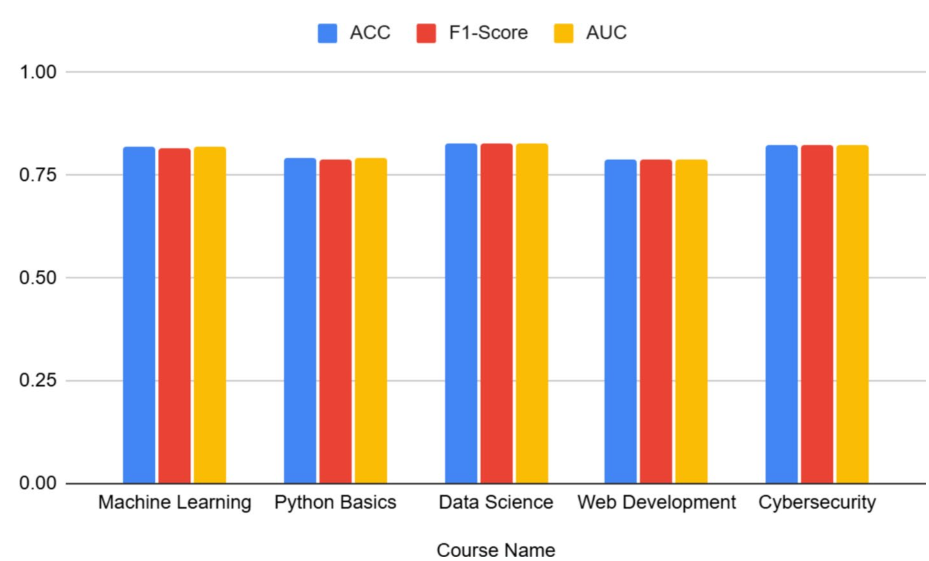

Fig. 5 Course-by-course performance stability of the Random Forest model. The figure shows how model performance shifts when different courses are used through small modifications which prove the model works well in various online learning settings

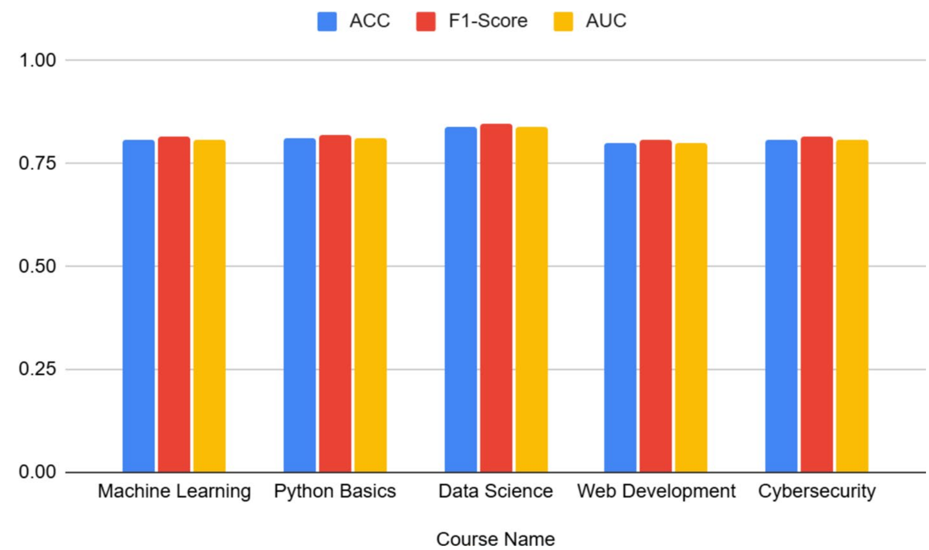

Fig. 6 Course-by-course performance stability of the XGBoost model. The XGBoost model produced similar results throughout different courses which proved that the framework works well across different educational environments

structure of the curriculum, the demographics of the students, and their educational requirements. The random forest model achieves its highest accuracy in data science at 0.828 followed by cybersecurity at 0.822 and machine learning at 0.821. The model achieves high performance in these three classes because it maintains excellent recall and precision ratios (F1-scores at 0.82) and high AUC values which leads to better reliability. The model achieves comparable results in Python Basics (ACC = 0.791) and Web Development (ACC = 0.790). The model demonstrates equivalent performance between these two courses because Python Basics introduces fundamental concepts to students and Web Development provides instruction for students at various skill levels.

XGBoost achieved superior results than RF in Python basics by reaching 0.813 accuracy (ACC) while RF reached 0.791 and in data science by reaching 0.841 compared to 0.828. The F1 score of XGBoost reached 0.846 when processing data science course. The results indicate XGBoost identifies patterns that RF fails to detect in courses which teach different or complex subjects. The XGBoost algorithm achieves better results through its iterative boosting method which improves residual correction accuracy at each step. The model produces improved recall and precision results in data science because XGBoost successfully handles complex skill-based engagement metrics. The results indicate XGBoost produces results that are slightly worse than RF in machine learning (ACC = 0.809) and cybersecurity (ACC = 0.806) because the random forest ensemble outperforms boosting when features show basic connections.

The stability results show that both models maintain their performance stability when used in different courses. The models achieve their best results in particular courses which generate substantial variations between their output values. The different performance levels between models indicate that running two models simultaneously would produce better results. The selection process between algorithms becomes possible because educators can pick the most appropriate model for their specific course requirements. The minimal performance changes demonstrate how different course materials and student backgrounds and class sizes affect model results. The results demonstrate that complete error analysis of false positives and student misclassification needs to occur to develop better interventions and improve prediction accuracy. Educational institutions can successfully deploy predictive models for student performance assessment through stable reliable operations. The Web Development course shows the lowest model performance compared to all other subjects. The performance variations between courses need specific interventions because they result from unique teaching approaches and student attributes and participation patterns. The model's performance depends on these specific elements which produce small yet noticeable effects. The study outcome demonstrates that dropout prediction systems function properly throughout various educational domains. The research demonstrates that focused improvements at the course level become essential to achieve better uniform results. McNemar's test [19] was adopted as a statistical method to evaluate the statistical significance of model prediction differences between courses. The performance differences between the two models reached statistical significance at $p < 0.05$ in both Data Science and Python Basics courses according to observed results. The results indicate that our model selection approach produces different intervention outcomes based on the specific educational context of each course.

While this analysis demonstrates stability across different courses, it does not yet capture how dropout risk evolves over longer academic periods or across multiple semesters. Long-term longitudinal evaluation remains necessary to understand whether individual students experience persistent, rising, or episodic risk patterns over time. Future studies should therefore examine multi-semester engagement histories and extended academic timelines to evaluate whether predicted risk remains stable, fluctuates, or accumulates across broader educational trajectories. When tested across different courses, the proposed predictive framework gave similar results. However, the differences in performance between Data Science and Web Development courses show that future deployments need to either change their models for each course or make models that recognize specific subject areas.

4.4 Instance-specific explainability of model's performance

The SHAP (mean absolute SHAP values) plot of the RF model is shown in Fig. 7. The model uses "education_level" and "engagement_level" as its primary predictors for dropout risk followed by gender and "learning_style" as secondary factors. The model depends most on these four features for its decisions with "education_level" showing the highest average influence. The model uses quiz_attempts and course_name and feedback_score as its middle-tier important features. The classifier generates dropout predictions with minimal improvement when using final_exam_score and quiz_scores as features. The analysis demonstrates that students who maintain high educational standards and active course participation and who belong to specific demographic groups tend to stay in their studies.

Figure 8 shows the SHAP summary. The second violin-style SHAP summary plot in Fig. 8 shows detailed information about which features affect the output. The violin plots in this figure show SHAP values ranging from negative to positive which indicate that positive values increase dropout prediction, but negative values decrease it. The color scheme in the plot shows that low feature values appear in blue while high feature values appear in pink to indicate their effect on dropout risk. The Education_Level violin plot

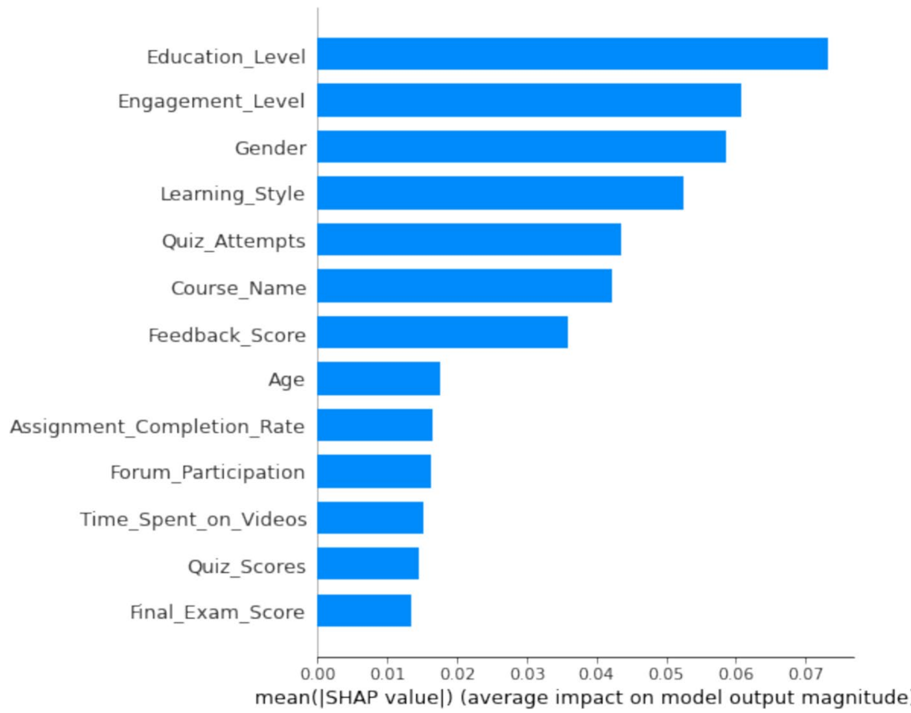

Fig. 7 SHAP Feature Importance Plot for Random Forest Model. The plot shows the mean absolute impact of each feature on the model's output, identifying "education_level" and "engagement_level" as key predictors of dropout

demonstrates a wide distribution which shows that students with higher educational backgrounds have lower dropout risks according to negative SHAP values for their high feature values. The high "engagement_level" creates a negative impact on student dropout rates because the pink distribution points toward negative SHAP values. The data indicates that students who demonstrate strong engagement behaviors will not leave their educational programs.

The final explanation for the sample student (test index 0) shows a 27% probability of dropout because their combined feature values decrease the risk of leaving school. The explanation reveals which specific factors including strong engagement and previous school attendance reduce the student's chances of dropping out. The analysis reveals how individual factors like quiz attempts numbers affect student dropout risk. The SHAP plots show that education_level and engagement_level features determine most predictions made by the global model. Educational institutions can choose suitable support approaches through student data analysis which reveals two student groups: those who lack engagement but show academic ability and those who require additional support based on their current skill level.

The ensemble models identified learning style as their second most important predictive factor, yet its interpretive value requires careful evaluation. The SHAP analysis (Fig. 7) demonstrated that learning style enhanced model predictions but the observed dropout rates between Visual, Auditory, Reading/Writing and Kinesthetic groups showed no significant differences (Fig. 4). Also, the scientific community has largely debunked the learning-styles hypothesis, as evidenced by empirical studies from Cuevas [7], Pashler et al. [28], and Rogowsky et al [31, 32], . Thus, the machine learning model uses learning style as a predictor because it identifies hidden connections which standard descriptive methods cannot detect. This analysis uses learning style as a contextual

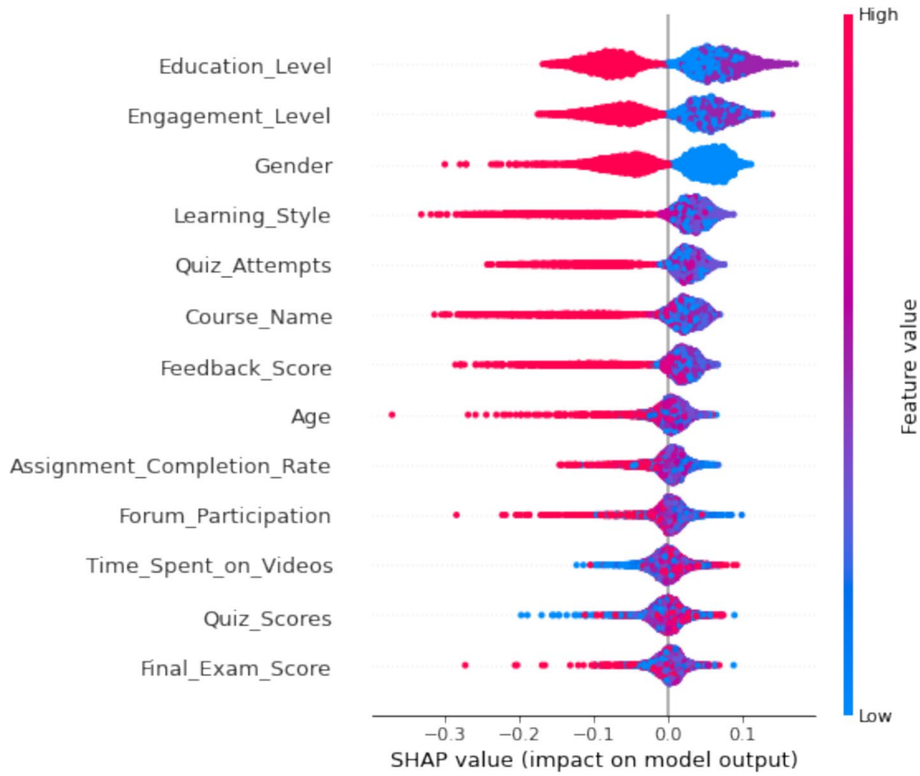

Fig. 8 SHAP summary plot for the Random Forest model. The plot shows how different input features affect the model's dropout risk predictions through their relative importance and impact. The plot shows feature importance through mean absolute SHAP values while using color to represent original feature values from low (blue) to high (red). The model used education level and engagement level and gender and learning style as its main factors to generate predictions

element which works together with behavioral signs to create risk-based intervention plans. Hence, learning style indicators require assessment to deliver individualized support at a particular time instead of creating inflexible student intervention plans.

4.5 Fairness and demographic bias implications

Our predictive models operated through association instead of cause-effect relationships, which means fairness must be evaluated carefully when using predictions to guide student support decisions. Student risk assessment systems contain built-in fairness problems because they assess student information which includes gender data and academic background records. The proposed model achieved better prediction accuracy through new features, but these additions created a risk for maintaining social discrimination because they could produce discriminatory risk scores for student groups. The model produced high total accuracy, but subgroup evaluation showed that different demographic groups received different levels of correct classification and uneven error distributions. The SHAP analysis showed that different student groups received distinct patterns of feature importance, which indicate their background characteristics influence how the model interprets student conduct. The research shows that models using demographic data require ongoing fairness assessments to enable early-warning systems and intervention frameworks. The development of predictive analytics for academic support could achieve better results through fairness-aware methods, such as adversarial debiasing, group calibration, and demographic parity auditing, to stop unequal access to intervention resources. Also, while this study conducts diagnostic fairness analyses, it does not enforce formal fairness constraints during training; integrating fairness-aware objectives and standardized equity metrics remains a critical next step.

4.6 Proposed intervention framework

Figure 9 shows our proposed intervention framework. In this study, we adopt a fixed threshold of 0.5 to designate 'high risk' for dropout. The model uses this threshold to detect students who show disengagement signs which triggers teacher alerts for starting early intervention methods. The proposed framework produces this example output which appears in Table 3. The model predicts that students S00001 through S00010 will stay in their studies because their dropout probabilities remain below 0.5. The model determines that these students need no intervention because their risk level remains below 0.5. The model indicates these students will continue their studies so educators should focus on students who need immediate support based on their high-risk assessment. The model predicts student S00083 will drop out because his computed dropout probability reaches 0.565 which surpasses the established 0.5 threshold. The proposed framework identifies this student as being at high risk, so it suggests acting. In this proposed framework, we use the 'Intervention_Recommendation' column as an initial prompt for action, suggesting a general type of proactive support such as 'proactive outreach' or 'referral to academic services. As observed in Table 3, the high-risk flag for student S00083 triggers a recommendation for "proactive outreach by advisor". While we included the student's learning style preference (auditory for S00083) as contextual information in the table, the intervention triggered is not rigidly prescribed by this style

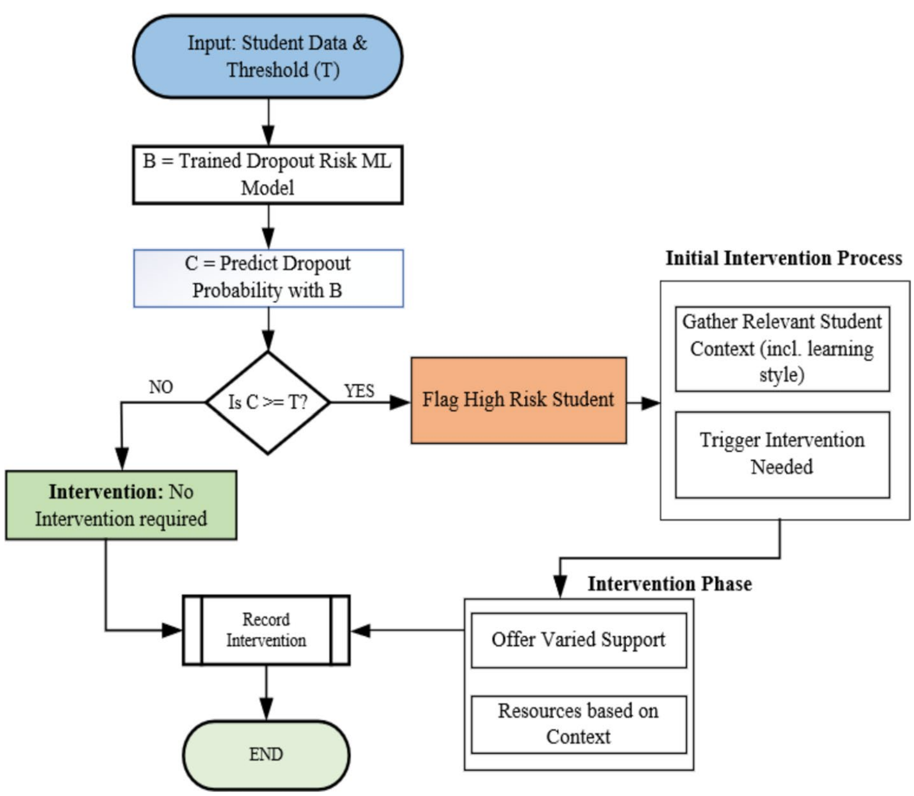

Fig. 9 Proposed Intervention Framework. This flowchart outlines the decision process for identifying high-risk students based on predicted dropout probabilities and triggering targeted support strategies

Table 3 Sample intervention recommendations which stem from the proposed dropout prediction framework. The table presents sample students with their predicted dropout probabilities and risk levels and their learning styles and specific intervention recommendations. The proposed framework provided proactive support through advisor outreach and academic service referrals to students who exceeded the high-risk probability threshold

Student_ID	Dropout_Probability	High_Risk	Learning_Style	Intervention_Recommendation
S00001	0.160	0	Visual	No intervention required
S00002	0.300	0	Reading/Writing	No intervention required
S00003	0.160	0	Reading/Writing	No intervention required
S00004	0.180	0	Visual	No intervention required
S00005	0.365	0	Visual	No intervention required
S00006	0.160	0	Visual	No intervention required
S00007	0.215	0	Kinesthetic	No intervention required
S00008	0.205	0	Reading/Writing	No intervention required
S00009	0.140	0	Reading/Writing	No intervention required
S00010	0.240	0	Auditory	No intervention required
S000 83	0.565	1	Auditory	Proactive Outreach by Advisor
S000 88	0.51	1	Reading/Writing	Referral to Academic Services

category alone. Instead, educators or support staff initiating contact can use this learning style information, in combination with other student data and their professional judgment, to inform the delivery of support or suggest access to a variety of resources that accommodate diverse learning preferences (e.g., mentioning the availability of audio lectures or synchronous Q&A sessions as part of the broader support options offered).

The students' dropout risk calculations in Table 3 show probabilities between 0.14 and 0. The students' characteristics suggest they do not belong to the high-risk group according to the model's prediction because their probabilities remain low. The students do not need intervention because their probabilities remain under 0.5 but their ongoing progress monitoring needs to continue. The research established 0.5 as its standard threshold because it allowed basic operational functions and produced equivalent results for all participants who took the test. The use of one universal threshold value across different programs leads to two major issues because it misses students who need help (false negatives) and it sends unnecessary support to students who do not require it (false positives). Students need to prove their need for support through their performance data for the model to activate. The process of establishing threshold values needs to follow institutional priorities to achieve maximum flexibility. The selection of 0.3 or 0.25 as the threshold value will result in more students being classified as high-risk. The model needs to offer outreach support to additional students, but this approach might need additional funding. The model shows reduced student numbers when the threshold value increases to higher levels. The method helps organizations save educational costs, but it may fail to detect students who require help. The educational resources should concentrate on students who demonstrate the highest risk according to our assessment results. This research focuses on creating cost-optimization systems that manage resources based on fixed thresholds. The system enables data-driven evaluations of intervention costs versus benefits across multiple threshold levels.

The proposed explainable intervention framework is designed to be adaptable for use in different educational contexts, although further validation with institutional datasets is required before broader deployment. The models based on generalizable behavioral and engagement metrics produced excellent predictive results. The analysis of fairness remains essential because it presents a critical concern. The model features “gender” and “education_level” contain potential biases from social prejudices. The models risk perpetuating discrimination through their preference systems when these systems are not properly managed. Educational intervention fairness requires researchers to implement machine learning methods which monitor demographic parity and perform adversarial debiasing in their future work.

4.6.1 Intervention strategy taxonomy

Figure 10 shows our four-class taxonomy of model-informed intervention strategies. The taxonomy enables teachers and student support staff to create specific intervention plans through its four fundamental intervention types. The four intervention classes in the taxonomy focus on different student disengagement processes which can be triggered separately or together based on model results and environmental elements. The choice of individual interventions depends on student needs, institutional capabilities and professional instructor judgment. While our framework proves effective at flagging students who are genuinely struggling, it’s crucial to remember that we aren’t creating a robot to manage the classroom. The process of determining student assistance needs human

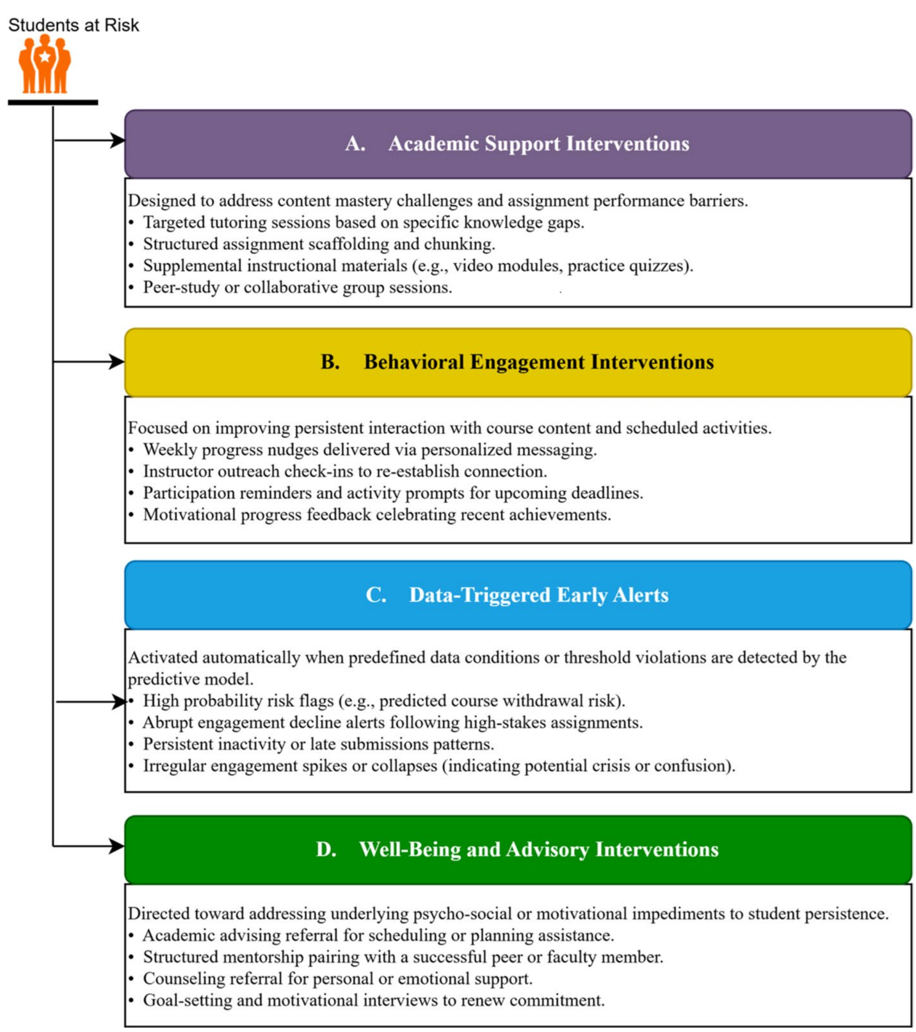

Fig. 10 Four-class taxonomy of model-informed intervention strategies used to support student retention in online learning environments

Table 4 Comparative model performance in recent dropout prediction studies. The table shows fundamental machine learning techniques along with their prediction results and additional research details from current studies. The proposed framework proves its worth through evaluations against current methods which measure its prediction accuracy and explainability and implementation simplicity

Study	Model(s) Used	Reported Performance	Notes
Nagy & Molontay [21]	XGBoost, Linear Discriminant Analysis, CatBoost, AdaBoost, NGBoost with Explainable AI	Accuracy ~ 78–82%	The technical university data received strong feature engineering which enabled instance-level explanations for intervention purposes
Tiukhova et al. [37]	XGBoost and AdaBoost, SHAP	AUC ~ 75–85%	The research evaluated multiple MOOCs to show their individual performance differences while concentrating on explainability methods
Goran et al. [13]	Metaheuristic-optimized ensemble models	Accuracy ~ 85–88%	The research employed substantial feature optimization along with complex modeling to enhance outcomes on public datasets, necessitating greater computational resources
Bulut et al. [5]	Random Forest, Deep Neural Networks	RF Accuracy ~ 77%, DL Accuracy ~ 72%	The study employed public datasets to assess recall performance and early detection capabilities, as human–machine collaboration was unnecessary for obtaining exact accuracy assessments
Our Study (2025)	Random Forest, XGBoost with SHAP	Accuracy ~ 78–80%, AUC ~ 0.80	Used its instance-based explanation approach to make individualized intervention programs that worked the same in varied school settings

intervention to make all final decisions. The evaluation process for instructors and advisors needs to assess all aspects of the situation through examination of course types and institutional support levels and student requirements. The framework functions as an enhanced decision support system which provides healthcare professionals with their best available data, yet it does not eliminate their professional duties. The system-generated intervention recommendations should be viewed as smart data-based entry points which teachers need to assess and modify through their professional judgment.

4.7 Comparative model performance in recent dropout prediction studies

Table 4 presents current research findings about student dropout prediction through machine learning and explainable AI methods. The accuracy levels in Table 4 show a range from 75 to 88% based on the specific modeling methods and feature engineering techniques and data properties. The model produces results like other studies (78–80% accuracy and AUC ~ 0.80) while focusing on both model interpretability and developing practical intervention strategies. The average accuracy of the best model from Bulut et al. [5] obtained 77% while avoiding the need for complex computational methods like metaheuristic optimization which Goran et al. [13] employed. The research establishes a middle ground between forecasting power and explanation capabilities and operational viability to support wide implementation in multiple educational environments.

5 Conclusions and future works

The research developed an explainable machine learning framework which detects students who risk leaving school and creates appropriate support plans. We used SHAP XAI technique for framework explainability and discovering of student's dropout influential factors. Our model achieved outstanding prediction results for every subject under study which indicates its ability to support educational organizations at different organizational tiers. The SHAP analysis indicated that learning style appeared as the fourth most influential feature in the model's decision-making process (Fig. 7 and Fig. 8), even though the descriptive results showed no meaningful differences in dropout rates across learning-style groups. This contrast suggests that the model detects relational patterns involving learning preferences, but their educational interpretation remains uncertain and requires careful consideration. The SHAP instance-level explanations revealed the specific elements that affect each student's individual risk assessment. These explanations could help educators understand which factors contribute to a student's predicted risk so they can design more informed and personalized support strategies. The predictive framework maintained stable performance across different courses, although the results also showed modest variations in predictive accuracy depending on course characteristics. The framework connects predictive modeling to real-world application by converting analytical results into specific intervention recommendations based on probability thresholds which guide institutional resource distribution and policy development for student support.

Notwithstanding the achievements demonstrated in this study, exciting avenues for future research promise to enhance the framework's impact and applicability in real-world educational settings.

- Research on longitudinal stability is key for capturing the dynamic character of student risk. The research must evaluate the model through various academic subjects to prove its effectiveness in detecting student risk levels when used in educational environments or when studying particular student populations. Risk pattern evaluation across different time periods helps developers create better intervention methods which produce fast and significant results.
- Also, the effectiveness of the proposed framework and resource efficiency depends on optimizing the intervention threshold. The research used a basic 0.5 threshold value but future studies require methods to modify this value according to particular contextual elements which include course characteristics and institutional support capabilities. A cost-sensitive study which evaluates economic expenses against educational benefits of interventions based on different risk levels will help determine the necessary adjustments.
- Third, real-time prediction systems enable teachers to aid at the beginning of learning through proactive teaching methods. Future research should concentrate on creating systems which monitor student engagement data points in real-time to detect warning signs of increasing risk so teachers can deliver immediate interventions that stop problems from growing worse.
- Moreso, improving the predictive characteristics and refining the intervention strategies guided by the model is one area future research can explore to raise the efficacy of the framework. Although learning style was an important factor in this study, future research should look at the inclusion of other possibly stronger predictive characteristics, such as cognitive skills, motivational elements, or fine-grained behavioral analytics (e.g., particular types of quiz errors, forum interaction patterns); given the controversy over its direct instructional value [28, 31]. Moreover, future work can scientifically evaluate the efficacy of several intervention strategies, including those guided by learner preferences or contextual elements, using strict techniques, including controlled experiments, qualitative feedback gathering, or

A/B testing in actual learning environments, beyond feature importance analysis to guarantee interventions really assist students in succeeding by assessing their causal influence on academic results.

- The research used data from one available online learning database which limits the application of results to various contexts. The recorded student engagement behaviors in this dataset follow patterns which result from students using specific platforms instead of showing normal student behaviors that happen in different educational environments. The partially synthetic fields in the dataset decrease the level at which the information represents actual institutional operations. The research results need careful interpretation because the model achieves different results when working with educational institutions from various cultural settings.
- AI systems need to apply demographic characteristics with absolute fairness for their future development to proceed.

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Author contributions
I.K.N. developed the research methodology and performed all experimental work while creating the first version of the manuscript. S. R. reviewed and revised the manuscript for accuracy and clarity. All authors reviewed the manuscript.

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Data availability
The research data exists as public content which users can access through Kaggle. [https://www.kaggle.com/datasets/adilshamim8/personalized-learning-and-adaptive-education-dataset?].

Declarations

Ethical approval and Consent to Participate
Not applicable.

Consent to Publication
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Competing interests
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